

[SHIELD] IGNISYL - ML Model Performance Report

Report Generated:	2025-12-26 09:55:42
Report Type:	Machine Learning Performance Analysis
Models Evaluated:	Isolation Forest, XGBoost, Autoencoder
Classification:	TECHNICAL

Executive Summary

This report provides comprehensive analysis of the IGNISYL machine learning models used for insider threat detection. The system employs an ensemble approach using Isolation Forest, XGBoost Classifier, and Autoencoder neural network. Current overall accuracy is 94.2% with a detection latency of 25ms.

1. Overall Performance Metrics

Metric	Value	Target	Status
Overall Accuracy	94.2%	> 90%	✓ PASS
False Positive Rate	5.00%	< 10%	✓ PASS
False Negative Rate	3.00%	< 5%	✓ PASS
Detection Latency	25ms	< 100ms	✓ PASS
Models Active	3	3	✓ PASS
Precision	92.8%	> 90%	✓ PASS
Recall	89.5%	> 85%	✓ PASS
F1 Score	91.1%	> 88%	✓ PASS

2. Individual Model Performance

Model	Accuracy	Precision	Recall	F1 Score	Latency
Isolation Forest	91.3%	89.7%	88.2%	88.9%	8ms

after_hours_access	0.065	■■■■■■■
sensitive_file_access	0.054	■■■■■■■
email_attachments	0.031	■■■

6. Recommendations for Model Improvement

- Consider increasing training data for threat samples to address class imbalance.
- Evaluate adding temporal features to capture time-series patterns in user behavior.
- Test LSTM or Transformer architectures for sequence-based anomaly detection.
- Implement periodic model retraining (weekly) to adapt to evolving threat patterns.
- Add ensemble voting weights optimization using genetic algorithms.
- Consider federated learning approach for privacy-preserving model updates.
- Implement A/B testing framework for safe model deployment.
- Add explainability features (SHAP values) for security analyst review.